Representation Theory and Locality in Celestial Holography

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Abstract

An introduction to celestial holography, leading up to some recent research on multi-collinear singular structures.

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1 Conventions

In Minkowski space we use the mostly-plus metric

$$\eta = \mathrm{diag}(-1, +1, \cdots, +1). \tag{1}$$

2 Motivation

TODO

2.1 AdS/CFT

TODO

2.2 Asymptotically Flat Space

We start in (3+1)-dimensional Minkowski space $(\mathbb{R}^4, \eta)$. Take the Minkowski metric in spherical coordinates

$$ds^2 = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \tag{2}$$

and introduce retarded and advanced null coordinates

$$(u, v) = (t - r, t + r) \tag{3}$$

to find

$$ds^2 = -dudv + \frac{1}{4}(u - v)^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{4}$$

Outgoing radiation lives at $v \to \infty$ for fixed (u, θ, ϕ), so in order to studying the asymptotic behavior of outgoing radiation (semiclassically, that of photons and gluons or even gravitons) we would like to compactify our spacetime, decorating it with a boundary on which this limit lives. Unfortunately our metric has a coordinate singularity in that limit. This can be seen directly from (4), or predicted on geometrical grounds - if we naively introduce a compactifying coordinate system with a boundary at infinity, many pairs of distinct points at infinity are infinitely far apart as measured by the flat bulk metric. So we know in advance that, if our program can be realized, any compactifying coordinate transformation will need to be accompanied by a Weyl transform. Identical comments apply to incoming radiation, which emerges from the region $u \to \infty$. When we take either of $u, v \to \infty$ while fixing the other and both of (θ, ϕ), we find coordinate divergences like $\mathcal{O}(u^2)$ or $\mathcal{O}(v^2)$, respectively. A natural spherically symmetric guess for an appropriate Weyl-transformed metric would be

$$\tilde{g}_{ab} = \Omega^2\eta_{ab} \text{ for } \Omega^2 = \frac{1}{P(u, v)} \tag{5}$$

where $P = \mathcal{O}(u^2)$ at fixed v and vice versa and where P is strictly positive for real coordinates. We choose in particular

$$\tilde{g}_{ab} = \Omega^2\eta_{ab} \text{ for } \Omega = \frac{4}{(1 + u^2)(1 + v^2)} \tag{6}$$

so that ds^2 is finite when one or both of u, v goes to $\pm\infty$. Now we play the compactification game, showing that the Weyl-transformed manifold $(\mathbb{R}^4, \tilde{g})$ lives naturally as an open submanifold in some larger manifold, then using the latter to identify a boundary for $(\mathbb{R}^4, \tilde{g})$. Concretely, we compactify u, v by passing them through an inverse tangent function, then linearly recombining them back into compactifed T, R coordinates as

$$\begin{aligned} T &= \tan^{-1} v + \tan^{-1} u, \\ R &= \tan^{-1} v - \tan^{-1} u \end{aligned} \tag{7}$$

In Minkowski space we have $u, v \in (-\infty, \infty)$ with $u \leq v$ but otherwise unconstrained, or equivalently in our new coordinates

$$\begin{aligned} -\pi &< T + R < \pi, \\ -\pi &< T - R < \pi, \\ R &\geq 0. \end{aligned} \tag{8}$$

That is, the natural domain of (T, R) looks like Figure 4 of [**?**]. Then our newly tamed and Weyl-transformed metric is

$$d\tilde{s}^2 = -dT^2 + dR^2 + \sin^2 R(d\theta^2 + \sin^2\theta d\phi^2) \tag{9}$$

which is exactly the metric on the *Einstein static universe* $\mathbb{R} \times S^3$ where the timelike coordinate T runs along $\mathbb{R}$ and where R plays the role of the (generalized) azimuthal angle on S^3. To help visualize this, we can repeat the same story in $(\mathbb{R}^2, \eta)$, where our initial and final metrics are

$$ds^2 = -dt^2 + dr^2 \xmapsto{\text{Weyl}} d\tilde{s}^2 = -dT^2 + dR^2 \tag{10}$$

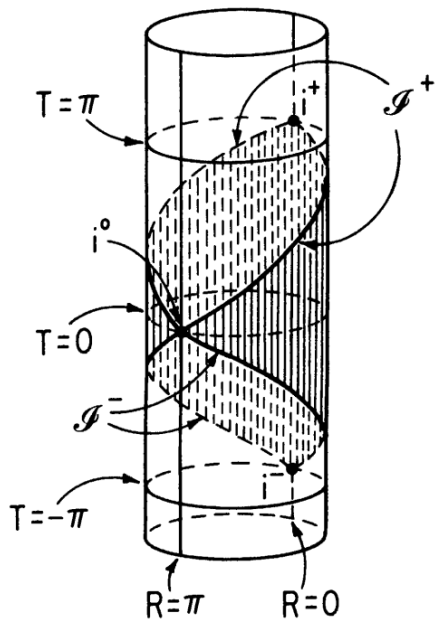

Figure 1: Embedding Weyl-transformed Minkowski space $(\mathbb{R}^2, \tilde{g})$ inside of the Einstein static universe $\mathbb{R} \times S^1$.

but where for lack of angular coordinates, to cover all of $\mathbb{R}^2$ we must take $(u, v) = (t - x, t + x)$. With this we have embedded $(\mathbb{R}^2, \tilde{g})$ into the (1+1)-dimension Einstein static universe $\mathbb{R} \times S^1$. The result is nicely illustrated by [**?**] Fig. 11.1, which we reproduce here in Fig. 1. Notice that in the latter, our embedding's S^1 factor leads us to identity $R \sim R + 2\pi$, hence dropping the constraint $R \geq 0$. In terms of our original coordinates, identifying the two poins $R = \pm\pi$ means that the spacelike asymptotic directions $x \to \pm\infty$ correspond to the same bound on the conformally compactified boundary. One might ask if this is strictly necessary - couldn't we further unwind the Einstein static universe under the universal covering map

$$\mathbb{R}^2 \to \mathbb{R} \times S^1, \quad (T, R) \mapsto (T, [R \sim R + 2\pi]) \tag{11}$$

and still have a perfectly good compactification of $(\mathbb{R}^2, \eta)$? Certainly, but this trick breaks above two dimensions since the n-sphere S^n is already simply connected for $n \geq 2$. In $3 + 1$ dimensions with Einstein static universe metric (9) for example, the coordinates $(T, -R, \theta, \phi)$ and $(T, R, \theta + \pi, \phi)$ correspond to the same point, so we can drop the coordinate constraint $R \geq 0$ without changing anything. Similar arguments hold in other dimension, so we have the following

Theorem There exists a conformal isometry of Minkowspace space $(\mathbb{R}^4, \eta)$ into the open region O of the Einstein static universe $(\mathbb{R} \times S^3, \tilde{g})$ given by the coordinate restriction $T \pm R \in (-\pi, \pi)$. A final notable feature of this story is that spacelike infinity $(u, v) \to (-\infty, \infty)$ corresponds in the Einstein static universe to setting $(T, R) = (0, \pi) \equiv (0, \pm\pi)$, *i.e.*, to the *one-dimensional (not three-dimensional) submanifold* $\mathbb{R} \times \{\text{south pole}\}$. This meas that, in the carton depiction 2 (lifted directly from [**?**] Fig. 11.2) is slightly misleading: not only do future and past timelike infinities $i^{\pm}$ (equivalently, $R = 0$ and $T = \pm\pi$) correspond to single points, so too does spacelike infinity i^0 (equivalently, $R \in \pi\mathbb{Z}$).

Intuitively, we want to call a spacetime *asymptotically flat* if we can conformally compactify it in a similar way, though with two dropped conditions. First, given the existence of stable massive objects in physically reasonable spacetimes, we won't require flatness as we go off in timelike directions, only spacelike and null. Second, we won't require that the Weyl-transformed metric have any smoothness properties at spacelike infinity, for reasons we'll see shortly.

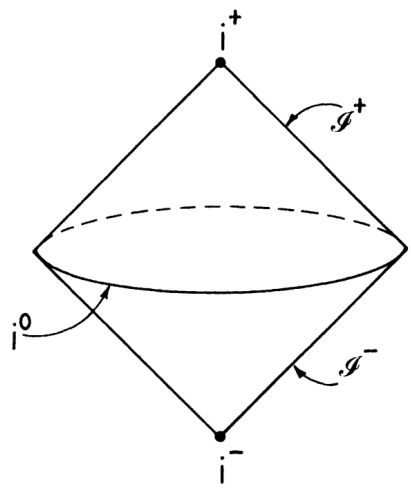

Figure 2: The region $\bar{O}$ of the Einstein static universe, represented as two null cones joined at the base.

2.3 Troubles with de Sitter

TODO

3 Kinematics of The Celestial Dual

Though we have discussed generally asymptotically flat spacetimes, we now restrict our attention to flat Minkowski space. We recall the well-known group isomorphism between the Lorentz group in signature $(d+1,1)$ and the Euclidean conformal group in d dimensions. At $d=2$ this becomes

$$SO(3,1) \simeq \mathrm{Conf}(2,0). \tag{12}$$

Here and elsewhere, when we refer to a Lie group we will always mean the its connected component containing the identity. Properly speaking we are therefore referring to the proper orthochronous Lorentz group $SO^{\uparrow}(3,1)$. On the other side of our isomorphism we are excluding orientation-reversing transformations; using the standard complex-analytic language we are equivalently restricting to holomorphic functions. Also as usual, the Euclidean conformal group in d dimensions should be thought of not as a group of automorphisms of real Euclidean space $\mathbb{R}^d$, but of its conformal compactification S^d. TODO: Implement this on the Lie-algebra level?

Inspired by the symmetry-matching game we played earlier for AdS/CFT, taking this isomorphism seriously leads one to wonder whether physics in Minkowski space might have a dual 2D conformal field theory. If we want this to lift to more general asymptotically flat spacetimes and have a shot and describing bulk quantum gravity, the dictionary ought to be built from bulk ingredients that make sense in such a theory.

The simplest candidate is the S-matrix, and so far this appears to be exactly the right object to use. An n-point amplitude has functional dependence

$$\mathcal{A} \sim \mathcal{A}_n\left(p_1, \sigma_1, n_1; \ldots; p_n, \sigma_n, n_n\right) \tag{13}$$

where, following the notation of [**?**], the ith particle has momentum p_i, spin/helicity labeled by σ_i, and flavor indices given by n_i. The idea behind the celestial dictionary is to parametrize our momenta and apply an appropriate change of basis after which the RHS to produce an object which transforms under Lorentz transformations (equivalently, under global conformal transformations) like the n-point vacuum correlator of a list of global conformal primaries. That is, we want

$$\mathcal{A}_n\left(p_1, \sigma_1, n_1; \ldots; p_n, \sigma_n, n_n\right) \xmapsto{\text{change of basis?}} \langle \mathcal{O}_{h_1,\bar{h}_1}(z_1,\bar{z}_1) \cdots \mathcal{O}_{h_n,\bar{h}_n}(z_n,\bar{z}_n)\rangle \tag{14}$$

where the RHS is to be computed in some dual theory on the celestial sphere S^2. Now the whole game of scattering is that amplitudes can be computed as matrix elements between a set of m incoming and k outgoing states for $m+k=n$.

$$\mathcal{A}_n\left(p_1,\sigma_1,n_1;\ldots;p_n,\sigma_n,n_n\right)=\left\langle p_1^{\text{in}},\sigma_1^{\text{in}},n_1^{\text{in}};\ldots;p_m^{\text{in}},\sigma_m^{\text{in}},n_m^{\text{in}};\right|S\left|p_1^{\text{out}},\sigma_1^{\text{out}},n_1^{\text{out}};\ldots;p_k^{\text{out}},\sigma_k^{\text{out}},n_k^{\text{out}}\right\rangle \tag{15}$$

By construction these transform under Lorentz as the tensor product of single-particle states, so it suffices to find an appropriate change of basis on single-particle basis states

$$|p,\sigma\rangle \xmapsto{\text{change of basis?}} \left|z_1,\bar{z}_1,h_1,\bar{h}_1\right\rangle \tag{16}$$

Before we work out the details of this change of basis, we make a slightly embarrassing admission: at the time of writing, it is not yet clear how to make this story work correctly when any of our bulk particles are massive. We can play the same representation-theoretic game, but the end result turns out not have the locality properties we would expect from a Euclidean n-point correlator. We will describe this problem more in detail later, but for now we restrict to the case that all off-shell momenta are massless.

3.1 Canonical and Non-Canonical Group Isomorphisms

In order to construct the celestial dictionary, we begin by spelling out the relevant group isomorphisms.

$$\begin{array}{ccc} SO^+(3,1) & \leftrightsquigarrow & SL(2,\mathbb{C})/\mathbb{Z}_2 \\ \updownarrow & & \updownarrow \\ \text{Conf}^+(S^2) & \leftrightsquigarrow & \text{Conf}^+(\mathbb{C}\cup\{\infty\}) \end{array}$$

We must first distinguish between the conformal automorphisms of the sphere from those the compactified plane. We write these groups respectively as $\text{Conf}(S^2)$ and $\text{Conf}(\mathbb{C}\cup\{\infty\})$, and use a superscript $^+$ to denote the connected component of any Lie group containing the identity (which for these groups is exactly the subgroup of orientation-preserving such maps). Of course the manifolds S^2 and $\mathbb{C}\cup\{\infty\}$ are conformally equivalent, but not canonically so. Equivalently, there are many many to endow S^2 with conformal coordinates $(z,\bar{z})$ so that the metric on S^2 is some local rescaling of $dz\wedge d\bar{z}$. In the commutative diagram above, we write a squiggly bidirectional arrow to indicate that $\text{Conf}(S^2)$ and $\text{Conf}(\mathbb{C}\cup\infty\}$ are non-canonically isomorphic groups. More quantitatively, we can describe the degree of non-canonicality by noting that any two conformal maps

$$\phi,\phi':\mathbb{C}\cup\{\infty\}\to S^2 \tag{17}$$

differ exactly by composition with a conformal transformation of S^2.

$$\phi'=\psi\circ\phi,\quad \psi=\phi'\circ\phi^{-1}:S^2\to S^2. \tag{18}$$

So the ambiguity is exactly the group of canonical automorphisms of S^2, which form a Lie group of real dimension 6.

Now for the top-most edge of our diagram. We recall that $SO^+(3,1)$ has a universal cover of degree 2 isomorphic to $SL(2,\mathbb{C})$, or equivalently that $SO^+(3,1)\simeq SL(2,\mathbb{C})/\mathbb{Z}_2$. The usual construction works by introducing the Pauli matrices

$$\begin{aligned} \left(\sigma^0_{\alpha\dot{\alpha}}\right)&=\begin{bmatrix}1&0\\0&1\end{bmatrix}, & \left(\sigma^1_{\alpha\dot{\alpha}}\right)&=\begin{bmatrix}0&1\\1&0\end{bmatrix}, \\ \left(\sigma^2_{\alpha\dot{\alpha}}\right)&=\begin{bmatrix}0&-i\\i&0\end{bmatrix}, & \left(\sigma^3_{\alpha\dot{\alpha}}\right)&=\begin{bmatrix}1&0\\0&-1\end{bmatrix} \end{aligned} \tag{19}$$

then building the natural mapping

$$\rho:\mathbb{R}^{3+1}\to\text{Her}(2,\mathbb{C}):=\{\text{Hermitian } 2\times 2 \text{ matrices}\} \tag{20}$$

given by

$$\rho(p)_{\alpha\dot{\alpha}} = \sigma^{\mu}_{\alpha\dot{\alpha}}\eta_{\mu\nu}p^{\nu} \equiv p_{\alpha\dot{\alpha}} \tag{21}$$

or in components

$$(p_{\alpha\dot{\alpha}}) = \begin{bmatrix} -p^0 + p^3 & p^1 - ip^2 \\ p^1 + ip^2 & -p^0 - p^3 \end{bmatrix}. \tag{22}$$

Now by direct computation we see that this obeys

$$\det(p_{\alpha\dot{\alpha}}) = \left(p^0\right)^2 - \left(p^1\right)^2 - \left(p^2\right)^2 - \left(p^3\right)^2 = -p \cdot p. \tag{23}$$

So the proper orthochronous Lorentz transformations on p correspond to the determinant- and Hermiticity-preserving linear maps on $\rho(p)$. The latter are exactly the conjugations

$$M \mapsto AMA^{\dagger} \text{ for } A \in SL(2, \mathbb{C}) \tag{24}$$

so any $A \in SL(2, \mathbb{C})$ has an associated proper orthchronous Lorentz transformation $\Lambda \in SO^{+}(3, 1)$ defined implicitly by the rule

$$A\rho(p)A^{\dagger} = \rho(\Lambda p) \text{ for all } p \in \mathbb{R}^{3+1}. \tag{25}$$

This mapping $A \mapsto \Lambda$ is exactly the projection map establishing $SL(2, \mathbb{C})$ as the double cover of $SO^{+}(3, 1)$. But we made a non-canonical choice: we chose (σ_{μ}) as a real basis for the set of all Hermitian 2×2 matrices satisfying the determinant condition (23). But this choice is non-unique; with a bit of thought, one can see that some other basis $(\hat{e}_{\mu})$ has this property iff

$$\hat{e}^{\mu} = \Lambda^{\mu}{}_{\nu}\hat{e}^{\nu} \tag{26}$$

for Λ a Lorentz transformation. Again, we see that the degree of ambiguity is parametrized by a Lie group of real dimension 6.

On the other hand, the isomorphisms indicated by solid bidirectional arrows are canonical. Along the rightmost edge of our diagram, we recall that the the conformal automorphisms of $\mathbb{C} \cup \{\infty\}$ connected to the identity are exactly the fractional linear transformations

$$z \mapsto \frac{az + b}{cz + d} \tag{27}$$

for $a, b, c, d \in \mathbb{C}$ satisfying WLOG $ad - bc = 1$. So to any special linear matrix we can associate such a mapping as

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \left(z \mapsto \frac{az + b}{cz + d}\right). \tag{28}$$

This group homomorphism is 2:1, with $\pm M$ having the same image, hence induces a canonical isomorphism between $\mathrm{Conf}^{+}(\mathbb{C} \cup \{\infty\})$ and $SL(2, \mathbb{C})/\mathbb{Z}_2$.

The canonical isomorphism between $SO^{+}(3, 1)$ and $\mathrm{Conf}(S^2)$ is best described in more geometrical terms. We associate to any nonzero null vector $p = (p^0, \vec{p}) \in \mathbb{R}^{3+1}$ the normalized future-directed null vector $\hat{p}$ by taking

$$p = p^0\hat{p} \implies \hat{p} = \left(1, \frac{\vec{p}}{p^0}\right) \text{ where } \frac{\vec{p}}{p^0} \in S^2 \tag{29}$$

The unit vectors $\hat{p}$ correspond naturally to points on S^2, and they get moved around if we replace p with Λp. So any Lorentz transformation $\Lambda \in SO(3, 1)$ induces an automorphism (which one can convince oneself is conformal) of S^2 given by

$$\hat{p} \mapsto \widehat{\Lambda p}. \tag{30}$$

This map is well-defined, as it doesn't depend on how at intermediate steps we lift $\hat{p}$ to p. It also illustrates exactly why the correct isomorphism is between the proper orthochronous Lorentz Lorentz transformations $\Lambda \in SO^{+}(3, 1)$ and orientation-preserving conformal transformations $f \in \mathrm{Conf}^{+}(S^2)$: any Lorentz transformation which isn't proper orthochronous must be a proper orthochronous one composed with the PT reflection $x^{\mu} \mapsto -x^{\mu}$, but such a transformation corresponds to the identity transformation on S^2. We see similarly that any

Lorentz transformation built from only one of P, T is orientation-reversing on S^2. So we can lift to a slightly more detailed homomorphism from Minkowski-norm-preserving linear transformations to not-necessarily-orientation-preserving conformal maps

$$\pi : O(3,1) \to \mathrm{Conf}(S^2). \tag{31}$$

The domain has four connected components corresponding to whether a transformation includes P and T, the codomain has two corresponding to whether a transformation flips orientation, and the map is 2:1 with kernel PT.

Now we want to parametrize the null external momenta appearing in our scattering amplitude in a way which plays nicely with these isomorphisms. Concretely, we want to put conformal coordinates $(z, \bar{z})$ on $S^2 \simeq \{\text{normalized null momenta}\}$. For reasons just described this will require some arbitrary choices, but we know that z should transform under a fractional-linear mapping when p transforms under a linear one, and we know that any two momenta differing by overall scaling should correspond to the same point on S^2. We therefore follow [**?**] and take

$$z = \frac{p^1 + ip^2}{p^0 + p^3} \tag{32}$$

From this we see immediately that a spatial rotation about the bulk x^3-axis and boosts along the same corresponding on the boundary to rotation and dilatations centered at the origin. Equivalently, this corresponds to putting coordinates on S^2 via a stereographic projection onto the x^1x^2-plane.

After some algebraic fiddling modulo the relation $p_\mu p^\mu = 0$ we find

$$\left(\frac{p^1}{p^0}, \frac{p^2}{p^0}, \frac{p^3}{p^0}\right) = \left(\frac{z+\bar{z}}{1+z\bar{z}}, \frac{-i(z-\bar{z})}{1+z\bar{z}}, \frac{1-z\bar{z}}{1+z\bar{z}}\right). \tag{33}$$

So up to an arbitrary choice of normalization (which we make slightly differently than we did above), we decompose

$$p(\omega, z, \bar{z}) = \omega\hat{p}(z, \bar{z}) = \omega\left(1 + z\bar{z}, z + \bar{z}, -i(z - \bar{z}), 1 - z\bar{z}\right). \tag{34}$$

3.2 Mellin-Transformation Diagonalizes Boosts

Now given a single-particle state of momentum $p = \omega\hat{p}(z, \bar{z})$ and helicity H, we can write

$$|p, H\rangle \equiv |\omega, z, \bar{z}, H\rangle. \tag{35}$$

We noted above that our goal is to produce from this ket another that transforms under global conformal transformations "like a global conformal primary." What exactly does this mean? Algebraically, we know [**?**] that under a global transformation

$$z \mapsto z' = \frac{az+b}{cz+d}, \quad ad - bc = 1 \tag{36}$$

a global conformal primary in a 2D CFT with conformal weights $(h, \bar{h})$ transforms as

$$\mathcal{O}'_{h,\bar{h}}(z', \bar{z}') = \left(\frac{dz'}{dz}\right)^{-h}\left(\frac{d\bar{z}'}{d\bar{z}}\right)^{-\bar{h}}\mathcal{O}_{h,\bar{h}}(z, \bar{z}) = (cz+d)^{-2h}(\bar{c}\bar{z}+\bar{d})^{-2\bar{h}}\mathcal{O}_{h,\bar{h}}(z, \bar{z}). \tag{37}$$

In terms of Lie algebra elements, $\mathcal{O}_{h,\bar{h}}(z, \bar{z})$ simultaneously diagonalizes the subgroup of $\mathrm{Conf}(\mathbb{C} \cup \{\infty\})$ which leaves $z, \bar{z}$ fixed, *i.e.*, the subgroup generated by rotations and dilatations centered at $(z, \bar{z})$. Inserting our operators at the origin for ease of notation (the general case can be recovered by conjugation by translation), under a rotation $z' = e^{i\theta}z$ we have $(a, b, c, d) = (e^{i\theta/2}, 0, 1, e^{-i\theta/2})$ hence

$$\mathcal{O}'_{h,\bar{h}}(0,0) = e^{-i\theta(h-\bar{h})}\mathcal{O}_{h,\bar{h}}(0,0) \equiv e^{-i\theta s}\mathcal{O}_{h,\bar{h}}(0,0). \tag{38}$$

while under under an origin-centered dilatation $z' = e^{\Omega}z$ we have

$$\mathcal{O}'_{h,\bar{h}}(0,0) = e^{-\Omega(h+\bar{h})}\mathcal{O}'_{h,\bar{h}}(0,0) \equiv e^{-\Omega\Delta}\mathcal{O}'_{h,\bar{h}}(0,0) \tag{39}$$

So as usual we identify the combinations

$$(\Delta, s) = \left(h + \bar{h}, h - \bar{h}\right) \tag{40}$$

as the conformal scaling weight and spin. By conjugation under translation, we see that in general $\mathcal{O}_{h,\bar{h}}(z,\bar{z})$ simultaneously diagonalized rotations around and dilatations about $(z,\bar{z})$. To match this in bulk Lorentz language, we need a $p(z,\bar{z})$-dependent ket which simultaneously diagonalizes rotation about and boosts along $p(z,\bar{z})$. We have already accomplished half of this: under a spatial rotation by θ about the spatial part of p, precisely by the definition of helicity we have

$$|\omega, z, \bar{z}, H\rangle \xmapsto{R_z(\theta)} e^{-i\theta H} |\omega, z, \bar{z}, H\rangle . \tag{41}$$

So we have already diagonalized boundary rotations centered at $(z,\bar{z})$ and identify the bulk helicity H with the boundary conformal spin s. We have not yet diagonalized boosts along p, under which our ket transforms as

$$|\omega, z, \bar{z}, H\rangle \xmapsto{B_z(\beta)} \left|e^{\beta}\omega, z, \bar{z}, H\right\rangle \tag{42}$$

where β is the rapidity of the boost. We can reverse-engineer what integral transform does the job: we define a new ket related to the old one by a Mellin-transform with respect to the scaling parameter ω.

$$|\Delta, z, \bar{z}, s\rangle = \int_0^\infty \frac{d\omega}{\omega} \omega^{\Delta} |\omega, z, \bar{z}, H = s\rangle . \tag{43}$$

Then under a boost we have

$$\begin{aligned} |\Delta, z, \bar{z}, s\rangle &\xmapsto{B_z(\beta)} \int_0^\infty \frac{d\omega}{\omega} \omega^{\Delta} \left|e^{\beta}\omega, z, \bar{z}, H = s\right\rangle \\ &= \int_0^\infty \frac{d\left(e^{-\beta}\omega\right)}{\left(e^{-\beta}\omega\right)} \left(e^{-\beta}\omega\right)^{\Delta} \left|e^{\beta}\left(e^{-\beta}\omega\right), z, \bar{z}, H = s\right\rangle \\ &= e^{-\beta\Delta} \int_0^\infty \frac{d\omega}{\omega} \omega^{\Delta} |\omega, z, \bar{z}, H = s\rangle \\ &= e^{-\beta\Delta} |\Delta, z, \bar{z}, s\rangle . \end{aligned} \tag{44}$$

TODO: Describe why these satisfy the right highest-weight condition to be conformal primaries. So far our argument is consistent with them being descendants.

Finally, we can write down our proposed celestial dictionary. Given a bulk scattering amplitude between n on-shell external particles each with null momentum $p = \omega\hat{p}(z,\bar{z})$, we integrate out each ω by a Mellin transform to product an object which, at least kinematically, behaves like an n-point correlator in a 2D CFT.

$$\langle \mathcal{O}_{h_1,\bar{h}_1}(z_1,\bar{z}_1) \cdots \mathcal{O}_{h_n,\bar{h}_n}(z_n,\bar{z}_n)\rangle = \left(\prod_{i=1}^{n} \frac{d\omega_i}{\omega_i} \omega_i^{\Delta_i}\right) \mathcal{A}_n\left(\omega_1\hat{p}(z_1,\bar{z}_1), H_1, n_1; \ldots; \omega_n\hat{p}(z_n,\bar{z}_n), H_n, n_n\right) . \tag{45}$$

From this we get our first concrete prediction about the putative dual CFT. For each special of bulk particle with spin s, the boundary theory's spectrum includes a continuous family of conformal primaries with conformal weights

$$\left(h, \bar{h}\right) = \left(\frac{\Delta + s}{2}, \frac{\Delta - s}{2}\right) . \tag{46}$$

parametrized by the continuum of possible values of Δ. So far it is not clear exactly what range of Δ's should be allowed. One principled statement we can make is that, if we run through this argument in position space for bulk dimension $(d+1,1)$, the conformal primary wavefunctions with weights in the principal series [**?**]

$$\Delta \in \frac{d}{2} + i\mathbb{R} \tag{47}$$

form a complete basis of normalizable solutions to the free wave equation.

3.3 Massive Particles

TODO

4 Dynamics of the Celestial Dual

4.1 OPEs Organized by Symmetry

One of the miracles of 2D CFTs is that, by repeatedly taking operator product expansions (OPEs), one can write the n-point correlator of an arbitrary collection of non-coincident operator insertions in terms of sums and products of lower-point functions. As 2pt and 3pt functions are forced by symmetry up to a finite number of constants, this process almost entirely constrains the dynamics of our theory.

TODO: Write this more clearly, exactly what is left unfixed by OPEs?

On the boundary, computing an OPE means considering a correlator

$$\langle A(z_1, \bar{z}_1) B(z_2, \bar{z}_2) \cdots \rangle \tag{48}$$

where we take $(z_1, \bar{z}_1)$ to be asymptotically close to $(z_2, \bar{z}_2)$, and $\cdots$ hides a finite number of other insertions some finite distance from $(z_2, \bar{z}_2)$. Then inside of our correlator we can replace

$$A(z_1, \bar{z}_1) B(z_2, \bar{z}_2) = \sum_{n, \bar{n} \in \mathbb{Z}} \frac{\{AB\}_{n,\bar{n}}(z_2, \bar{z}_2)}{z_{12}^n \bar{z}_{12}^{\bar{n}}} \tag{49}$$

where

$$(z_{ij}, \bar{z}_{ij}) := (z_i - z_j, \bar{z}_i - \bar{z}_j) \tag{50}$$

and where the coefficient fields $\{AB\}_{n,\bar{n}}$ are independent of the remaining insertions, are nonsingular at $(z_2, \bar{z}_2)$, and vanish for sufficiently large $n, \bar{n}$. This series converges exactly when $(z_1, \bar{z}_1)$ is closer to $(z_2, \bar{z}_2)$ then any other point in the correlator is. Knowing the OPE between A and B is equivalent to knowing all of the coefficients $\{AB\}_{n,\bar{n}}$.

Now we change notation slightly. We write our operators at $\mathcal{O}_i$, with conformal weights $(h_i, \bar{h}_i)$. By translation covariance we can also expand $(z, \bar{z})$ about the origin rather than $(z_1, \bar{z}_1)$ about $(z_2, \bar{z}_2)$, writing slightly more neatly

$$\mathcal{O}_1(z, \bar{z}) \mathcal{O}_2(0, 0) = \sum_{n, \bar{n} \in \mathbb{Z}} \frac{\{\mathcal{O}_1 \mathcal{O}_2\}_{n,\bar{n}}(0, 0)}{z^n \bar{z}^{\bar{n}}}. \tag{51}$$

Covariance under the conformal group strongly constrains the form of the RHS. For one, the LHS and RHS must transform in the same way under conformal transformations. The LHS has conformal weights $(h_1 + h_2, \bar{h}_1 + \bar{h}_2)$

$$\mathcal{O}_1(z, \bar{z}) \mathcal{O}_2(0, 0) \sim (h_1 + h_2, \bar{h}_1 + \bar{h}_2) \tag{52}$$

while $z^n \bar{z}^{\bar{n}}$ has conformal weights

$$z^n \bar{z}^{\bar{n}} \sim (-n, -\bar{n}) \tag{53}$$

so we can read off the conformal weights of the coefficients as

$$\{\mathcal{O}_1 \mathcal{O}_2\}_{n,\bar{n}} \sim (h_1 + h_2 - n, \bar{h}_1 + \bar{h}_2 - \bar{n}). \tag{54}$$

We can make more progress by working in the basis for local operators given by conformal primaries $\mathcal{O}_P$ and their descendants $\partial^n \bar{\partial}^{\bar{n}} \mathcal{O}_P$. Denoting by C_{12}^P the OPE coefficient of the primary $\mathcal{O}_P$ and by $\beta_{12n\bar{n}}^P$ the relative coefficient of the $(n, \bar{n})$th descendant (so by construction $\beta_{1200}^P = 1$), we have

$$\mathcal{O}_1(z, \bar{z}) \mathcal{O}_2(0, 0) = \sum_P C_{12}^P \sum_{n, \bar{n}} \frac{\beta_{12n\bar{n}}^P}{z^{h_1 + h_2 - h_P} \bar{z}^{\bar{h}_1 + \bar{h}_2 - \bar{h}_P}} \partial_n \bar{\partial}_{\bar{n}} \mathcal{O}_P(0, 0). \tag{55}$$

This form of the OPE is manifestly covariant under translation, rotation, and scaling, but for totally general coefficients C, β does not yet enforce covariance under special conformal transformations. As it turns out [**?**], further enforcing covariant under SCTs (literally applying $[L_1, \cdot]$ and $[\bar{L}_1, \cdot]$ to both sides and equating the results term-by-term) recursively determinants the β coefficients, leaving only the C's as free data. In fact in a celestial CFT we can do better, since Lorentz symmetry embeds inside of a broader Poincaré group, and this extension on the boundary contrains even the leading coefficients C_{12}^P. More on this later.

4.2 OPEs from Bulk Scattering

We have just seen that boundary global conformal symmetry recursively determines the coefficients of descendants appearing in the OPE between any two primaries, so it suffices to compute just the OPE coefficients C_{12}^P between primaries. We also recall that each continuous family of boundary primaries corresponds to a bulk particle species. Now we make a very nice observation: from our celestial dictionary, we see that taking two insertions $(z_1, \bar{z}_1), (z_1, \bar{z}_2)$ asymptotically close corresponds to taking the corresponding bulk external momenta $p_i = \omega_i \hat{p}(z_i, \bar{z}_i)$ for $i = 1, 2$ asymptotically collinear.

Luckily for us, it is straightforward to see in diagrammatic language why (and when) bulk scattering amplitudes diverge in this limit. If we consider a scattering amplitude with external legs $1, 2$, and some other particle species 3 where $1 + 2 \to 3$ scattering has a nonzero amplitude, then a subset of diagrams contributing to $1 + 2 \to (n - 2)$ scattering will have an internal 3 leg forced by momentum conservation to carry momentum $p_1 + p_2$. In our parametrization, the associated propagator is (modulo convention-dependent constants)

$$\frac{1}{(p_1 + p_2)^2} = \frac{1}{2p_1 \cdot p_2} = \frac{1}{2\omega_1\omega_2 \hat{p}(z_1, \bar{z}_1) \cdot \hat{p}(z_2, \bar{z}_2)} = \frac{-1}{4(z_1 - z_2)(\bar{z}_1 - \bar{z}_2)}. \tag{56}$$

Not only does this explain why boundary coincident singularities make sense from a bulk scattering perspective, it gives us highly quantitative information:

- The primaries P appearing in the boundary $12 \to P$ OPE correspond to bulk particle species with a nonzero scattering amplitude against $1 + 2$, with OPE coefficient scaling with the amplitude for such a process.
- The boundary OPEs are most inverse-linear in z_{12} and $\bar{z}_{12}$.

Conversely, this hints at an analytic subtlety. So far we have been doing all of our complex analysis in a formalism which (at least typographically) treats $z, \bar{z}$ as independent variables. Said more concretely, a complex function which isn't holomorphic may nonetheless admit a local Laurent series involving powers of both z and $\bar{z}$

$$f : \mathbb{C} \to \mathbb{C}, \quad f(z) = \sum_{n, \bar{n} \in \mathbb{Z}} a_{n\bar{n}} z^n \bar{z}^{\bar{n}}. \tag{57}$$

This series expansion then defines a function of two complex variables by taking $z, \bar{z} \in \mathbb{C}$ to be independent. We recover f by working on the "real surface" where $z, \bar{z}$ are conjugates. This is a standard game to play when studying 2D CFTs. On the amplitudes side, however, it raises some concern.

Like OPEs in a 2D CFT, bulk amplitudes are productively studied by continuing off the real surface. Equivalently, if we use spinor helicity notation to factor our null momenta as

$$p_{\alpha\dot{\alpha}} \equiv p_\mu \sigma^\mu_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}} \tag{58}$$

it is illuminating to analytically continue to complex momenta by dropping the assumption that $\lambda, \tilde{\lambda}$ be componentwise complex conjugates of one another. But then we have have variously

$$p_1 \cdot p_2 \sim (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) \sim (\lambda_1 \cdot \lambda_2)(\tilde{\lambda}_1 \cdot \tilde{\lambda}_2). \tag{59}$$

So taking $p_1 \cdot p_2 \to 0$ can be accomplished by taking $z_1 - z_2 \sim \lambda_1 \cdot \lambda_2 \to 0$, or $\bar{z}_2 - \bar{z}_2 \sim \tilde{\lambda}_1 \cdot \tilde{\lambda}_2 \to 0$, and these limits may or may not commute. Certainly from the bulk side it's often unclear the correct limit to take, and much of the amplitudes literature contents itself wih taking the holomorphic limit $z_1 - z_2 \sim \lambda_1 \cdot \lambda_2 \to 0$ with the

anti-holomorphic variables fixed. From this one can recover the opposite limit by symmetry, though recovering the simultaneous limit is more difficult.

In any case, if we are happy to worry just about the holomorphic limit we can use known properties of bulk collinear limits to infer the corresponding boundary OPEs in the same kind of limit. The strategy is as described above: any $1+2 \to 3$ channel in the bulk gives rise to a contribution $\mathcal{O}_3$ to the boundary $\mathcal{O}_1\mathcal{O}_2$ OPE with singularity $\sim \frac{1}{z_{12}}$, plus subleading terms corresponding on the boundary to descendants of $\mathcal{O}_3$. The structure of the bulk 3pt vertex then fixes the finer structure of the boundary OPE.

In the simplest example, we can consider tree-level pure-glue Yang-Mills [**?**]. Free of loop-level and non-perturbative corrections, our bulk has only two bulk particle species[1] So tree level $GG \to G$ bulk scattering channel in gives rise to the holomorphic OPE channel

$$\mathcal{O}_{\Delta_1,s_1}(z_1,\bar{z}_1)\mathcal{O}_{\Delta_2,s_2}(z_2,\bar{z}_2) \propto \frac{1}{z_{12}}\mathcal{O}_{\Delta_3,s_3}(z_3,\bar{z}_3) + (\text{descendants}) + (\text{other channels}) + (\text{reg}) \tag{60}$$

where $\mathcal{O}_{\Delta,s}$ for $s=\pm 1$ is the celestial dual to a gluon with helicity s. While a Mellin transform in a vacuum has Δ as a free variable, it turns out that bulk physics fixes Δ_3 in the above OPE. In more detail, a 3pt interaction vertex

$$V_{3\text{pt}} \sim (\partial_{m_1}\Phi_1)\,(\partial_{m_2}\Phi_2)\,(\partial_{m_3}\Phi_3) \quad m_1+m_2+m_3 \equiv m \tag{61}$$

has mass dimension $3+m$[2] This gives rise to a $1+2 \to 3$ scattering channel with amplitude scaling like p_3^m. Schematically, we then have a singularity... TODO: come back to this. Need an elementary argument. Anyway we know

$$\Delta_3 = \Delta_1 + \Delta_2 + d_V - 5. \tag{62}$$

In the case of pure-glue tree-level Yang-Mills, the authors of [**?**] use these techniques to determine the leading-order holomorphic singular between celestial positive-helicity gluons to be

$$\mathcal{O}^{+a}_{\Delta_1}(z_1,\bar{z}_1)\mathcal{O}^{+b}_{\Delta_2}(z_2,\bar{z}_2) = -\frac{if^{ab}{}_c}{z_{12}}B(\Delta_1-1,\Delta_2-1)\mathcal{O}^{+c}_{\Delta_!+\Delta_2-1}(z_2,\bar{z}_2) + (\text{descendants}) + (\text{reg}). \tag{63}$$

Then the recursive structure forced by covariance under special conformal transformations further fixes the descendant contributions and we have more precisely

$$\mathcal{O}^{+a}_{\Delta_1}(z_1,\bar{z}_1)\mathcal{O}^{+b}_{\Delta_2}(z_2,\bar{z}_2) = -\frac{if^{ab}{}_c}{z_{12}}\sum_{n=0}^{\infty}\bar{z}_{12}^n B(\Delta_1+n-1,\Delta_2-1)\bar{\partial}^n\mathcal{O}^{+c}_{\Delta_!+\Delta_2-1}(z_2,\bar{z}_2) + (\text{reg}). \tag{64}$$

Note that the answer is simplified because any holomorphic descendants would contribute only to the regular part. Similar results hold for generalizations to mixed-helicity gluons, to tree-level gravity, and to their combination into tree-level Einstein-Yang-Mills.

5 Multicollinear Limits

A corner of this picture which has been somewhat obscure until recently is the extension to multicollinear amplitudes. In the simplest case the idea is to take three off-shell momenta to all asymptotically parallel to one another and see what kinds of singularities arise. In more detail, to make the procedure well defined we must take p_1, p_2 asymptotically collinear, regularize by throwing out any singularities that arise, then take p_3 asymptotically collinear to the first two and see what singularities arise in this second limit. In other words we have $\hat{p}_i \cdot \hat{p}_j$ asymptotically small, but $\hat{p}_1 \cdot \hat{p}_2$ parametrically smaller than $\hat{p}_1 \cdot \hat{p}_3 \approx \hat{p}_2 \cdot \hat{p}_3$.

Prior to our work [**?**], multicollinear limits in celestial CFTs were mainly studied by computing multicollinear bulk amplitudes as just described and Mellin-transforming only at the end. This has a number of disadvantages:

[1] Our phrasing here is literal and intentional: under the standard definition of a particle species as a sufficiently small irrep of $SO^+(3,1)$ in our Hilbert space, a positive-helicity gluon cannot be transformed into a negative one. This point is often glossed over.

[2] A bosonic field with a standard kinetic term in D dimensions has mass dimension $\frac{D-2}{2}$, which in $3+1$ reduces to 1.

- The interaction between holomorphic and anti-holomorphic singularities is obscured and sometimes appears to rely on arbitrary choices.
- Amplitudes are hard to compute exactly, and it is often tractable to compute the leading collinear singularities but not subleading ones. Extracting multicollinear singularities directly often rely on recursive schemes (*e.g.*, [**?**]), which are scarce. Meanwhile, a priori it is possible for the leading multicollinear singularities to receive contributions from subleading collinear singularities.
- In bulk terms it is far from obvious how single- and multi-particle contributions to the multicollinear OPE constrain themselves. Concretely for example, the authors of [**?**] describe the contributions as essentialls independent. (TODO: Explain this better)

On the other hand, computing multicollinear OPEs in a 2D CFT is extremely natural, follows straightforward from pairwise OPE data, and involves no arbitrary decisions, essentially due to the highly constrained structure of complex analysis. The principle contributions of [**?**] were to

- Explain the principled boundary-first recipe for computing multicollinear celestial OPEs.
- Compute these directly from the bulk in a few of the rare theories were doing so is tractable by hand, and demonstrate agreement with the boundary calculation.
- Explain why, even if one doubts the existence or niceness of the putative dual CFT, agreement was guaranteed in advance. In some sense the right recipe for computing this multicollinear limits has no physical content, but instead is forced on us by the structure of complex analysis.
- Explain why the structure is similarly forced in terms of the boundary image of the enhancement of Lorentz symmetry to Poincaré.

5.1 The Recipe from the Boundary: Composite Products and Generalized Wick

TODO: Uniformize notation in this section.

Translating the bulk procedure just described into boundary terms, we want to

- Insert operators $\mathcal{O}_i(z_i, \bar{z}_i)$ for $i = 1, 2, 3$ inside of a correlator along arbitrary other non-coincident insertions.
- Take $(z_2, \bar{z}_2) \to (z_3, \bar{z}_3)$. At this stage one can include only holomorphic singularities, only anti-holomorphic ones, or both, as long as one does the same thing at later steps.
- Throw out the singularities from the previous step. As $z_{12}, \bar{z}_{12} \to 0$ we automatically lose any regular terms of order $z_{12}^n \bar{z}_{12}^{\bar{n}}$ where at least one of $n, \bar{n}$ is strictly positive, so in fact at this point we are extracting the $\mathcal{O}(1)$ term.
- Now take $(z_1, \bar{z}_1) \to (z_3, \bar{z}_3)$.

The result of the first three steps is a well-known object called the *composite* or *normal-ordered* product. In standard presentations of QFT the normal-ordered product of a string of coincident insertions is defined in a free theory (or by extension, perhaps in perturbation theory), and is computed by expanding in terms of creation and annihilation modes then reordering to place all annihilation operators on the right (up to signs depending on field statistics). In practice, inside of a correlation function this is supposed to be an equivalent product but with the singular parts subtracted off. In a general interacting theory this recipe ceases to make sense, but in a 2D CFT the structure of complex analysis gives a canonical way to accomplish this game of subtracting off singular parts. Assuming an OPE of the form[3]

$$A(z, \bar{z})B(w, \bar{w}) = \sum_{n,\bar{n}} \frac{\{AB\}_{n,\bar{n}}(w, \bar{w})}{(z-w)^n (\bar{z}-\bar{w})^{\bar{n}}} \tag{65}$$

extracting the value at the coincident limit $(z, \bar{z}) \to (w, \bar{w})$ but with the singular part stripped off means extracting the operator $\{AB\}_{0,0}(0, 0)$. So we define

$$:AB:(w, \bar{w}) := \{AB\}_{0,0}(w, \bar{w}). \tag{66}$$

[3]Here we use slightly simpler but equivalent notation to avoid a proliferation of indices.

This definition in terms of Laurent expansions is immediately equivalent to an integral prescription for computing the composite product, as directly from the Cauchy residue theorem we have

$$: AB : (w, \bar{w}) = \oint_w \frac{dz}{2\pi i} \frac{1}{z-w} \oint_{\bar{w}} \frac{d\bar{w}}{2\pi i} \frac{1}{\bar{z}-\bar{w}} A(z,\bar{z}) B(w,\bar{w}). \tag{67}$$

Here $\oint_w$ denotes a contour running along a positively-oriented circle around w, with a radius smaller enough not to enclose any other insertions appearing inside of the implied correlator but otherwise arbitrary, and similarly for $\oint_{\bar{w}}$.

Again mirroring the notation used for free fields, we denote by $\overbracket{A(z,\bar{z})B}(w,\bar{w})$ the part of the expansion containing exclusively singular parts. That is,

$$\overbracket{A(z,\bar{z})B}(w,\bar{w}) = \sum_{\max(n,\bar{n})>0} \frac{\{AB\}_{n,\bar{n}}(w,\bar{w})}{(z-w)^n(\bar{z}-\bar{w})^{\bar{n}}}. \tag{68}$$

Then it turns out that a similarly simple contour integral prescription fixes the singular part of the OPE $A : BC :$ in terms of the singular parts of the pairwise OPEs. Concretely, the generalized Wick theorem [**?**] tells us

$$\begin{aligned} &\overbracket{\mathcal{O}_1(z_1,\bar{z}_1) : \mathcal{O}_2}\mathcal{O}_3 : (z_3,\bar{z}_3) \\ &= \oint_{z_2} \frac{dz_2}{2\pi i} \frac{1}{z_{23}} \oint_{\bar{z}_2} \frac{d\bar{z}_2}{2\pi i} \frac{1}{\bar{z}_{23}} \left(\overbracket{\mathcal{O}_1(z_1,\bar{z}_1)\mathcal{O}_2}(z_2,\bar{z}_2)\mathcal{O}_3(z_3,\bar{z}_3) + \overbracket{\mathcal{O}_1(z_1,\bar{z}_1)\mathcal{O}_3}(z_3,\bar{z}_3)\mathcal{O}_2(z_2,\bar{z}_2) \right). \end{aligned} \tag{69}$$

Applying this to a theory with known pairwise OPEs lets us compute multiparticle OPEs directly with no need to compute involved multicollinear amplitudes in the bulk. For example, in (tree-level, matter-free, perturbative about flat space) Einstein gravity we have pairwise OPEs in the holomorphic limit [**?**]

$$G^{s_1}_{\Delta_1}(z_1,\bar{z}_1) G^{s_2}_{\Delta_1}(z_2,\bar{z}_2) = -\delta_{s_1+s_2\geq 0} \frac{\kappa}{2} \frac{\bar{z}_{12}}{z_{12}} B(\Delta_1 - s_1 + 1, \Delta_2 - s_2 + 1) G^{\min(s_1,s_2)}_{\Delta_1+\Delta_2}(z_2,\bar{z}_2) \tag{70}$$

Plugging this into the generalized Wick theorem (69) gives at leading order

$$\begin{aligned} &G^{s_1}_{\Delta_1}(z_1,\bar{z}_1) : G^{s_2}_{\Delta_2} G^{s_3}_{\Delta_3} : (z_3,\bar{z}_3) \\ &\quad \sim \frac{\kappa^2}{4} \frac{\bar{z}_{13}^2}{z_{13}^2} \delta_{s_1+s_2+s_3\geq 0} B(\Delta_1 - s_1 + 2, \Delta_2 - s_2 + 1, \Delta_3 - s_3 + 1) G^{\min(s_1,s_2,s_3)}_{\Delta_1+\Delta_2+\Delta_3}(z_3,\bar{z}_3) \\ &\qquad - \frac{\kappa}{2} \frac{\bar{z}_{13}}{z_{13}} \delta_{s_1+s_2\geq 0} B(\Delta_1 - s_1, \Delta_2 - s_2 + 1) : G^{\min(s_1,s_2)}_{\Delta_1+\Delta_2} G^{s_3}_{\Delta_3} : (z_3,\bar{z}_3) \\ &\qquad - \frac{\kappa}{2} \frac{\bar{z}_{13}}{z_{13}} \delta_{s_1+s_3\geq 0} B(\Delta_1 - s_1 + 1, \Delta_3 - s_3 + 1) : G^{s_2}_{\Delta_2} G^{\min(s_1,s_3)}_{\Delta_1+\Delta_3} : (z_3,\bar{z}_3). \end{aligned} \tag{71}$$

If we run through the procedure with all descendants in the pairwise OPEs (and straighten up notation slightly by writing in terms of $(h,\bar{h})$ rather than (Δ, s)) we get more precisely

$$\begin{aligned} &G^{s_1}_{\Delta_1}(z_1,\bar{z}_1) : G^{s_2}_{\Delta_2} G^{s_3}_{\Delta_3} : (z_3,\bar{z}_3) \\ &\quad = \frac{\kappa^2}{4} \frac{\bar{z}_{13}^2}{z_{13}^2} \delta_{s_1+s_2+s_3\geq 0} \sum_{m=0}^{\infty} \frac{\bar{z}_{13}^m}{m!} B(2\bar{h}_1 + 2 + m, 2\bar{h}_2 + 1, 2\bar{h}_3 + 1) \bar{\partial}^m G^{\min(s_1,s_2,s_3)}_{\Delta_1+\Delta_2+\Delta_3}(z_3,\bar{z}_3) \\ &\qquad - \frac{\kappa}{2} \frac{\bar{z}_{13}}{z_{13}} \delta_{s_1+s_2\geq 0} \sum_{m=0}^{\infty} \frac{\bar{z}_{13}^m}{m!} B(2\bar{h}_1 + 1 + m, 2\bar{h}_2 + 1) : \left(\bar{\partial}^m G^{\min(s_1,s_2)}_{\Delta_1+\Delta_2}\right) G^{s_3}_{\Delta_3} : (z_3,\bar{z}_3) \\ &\qquad - \frac{\kappa}{2} \frac{\bar{z}_{13}}{z_{13}} \delta_{s_1+s_3\geq 0} \sum_{m=0}^{\infty} \frac{\bar{z}_{13}^m}{m!} B(2\bar{h}_1 + 1 + m, 2\bar{h}_3 + 1) : G^{s_2}_{\Delta_2} \left(\bar{\partial}^m G^{\min(s_1,s_3)}_{\Delta_1+\Delta_3}\right) : (z_3,\bar{z}_3). \end{aligned} \tag{72}$$

If we take the general form (55), restrict to the celestial case of inverse-linear poles, and run again through the generalized Wick prescription (69), we find the maximally general result

$$
\begin{aligned}
&\mathcal{O}_1(z_1,\bar{z}_1):\mathcal{O}_2\mathcal{O}_3:(z_3,\bar{z}_3)\\
&\quad\sim\sum_{I,J}\frac{\gamma_{s_I}^{s_1,s_2}\gamma_{s_J}^{s_I,s_3}}{z_{13}^2}\bar{z}_{13}^{p_{12I}+p_{I3J}}B(2\bar{h}_1+p_{12I}+p_{I3J},2\bar{h}_2+p_{12I},2\bar{h}_3+p_{I3J})\mathcal{O}_J(z_3,\bar{z}_3)\\
&\qquad+\sum_{I,J}\frac{\bar{\gamma}_{s_I}^{s_1,s_2}\bar{\gamma}_{s_J}^{s_I,s_3}}{\bar{z}_{13}^2}z_{13}^{\bar{p}_{12I}+\bar{p}_{I3J}}B(2h_1+\bar{p}_{12I}+\bar{p}_{I3J},2h_2+\bar{p}_{12I},2h_3+\bar{p}_{I3J})\mathcal{O}_J(z_3,\bar{z}_3)\\
&\qquad+\sum_{I,J}\delta_{\bar{p}_{I3J},0}\frac{\gamma_{s_I}^{s_1,s_2}\bar{\gamma}_{s_J}^{s_I,s_3}}{z_{13}}\bar{z}_{13}^{p_{12I}-1}B(2h_1+2h_2-2,2h_3)\\
&\qquad\qquad\times\left[B(2\bar{h}_1+p_{12I}-1,2\bar{h}_2+p_{12I})-B(2\bar{h}_1+p_{12I}-1,2\bar{h}_2)\right]\mathcal{O}_J(z_3,\bar{z}_3)\\
&\qquad+\sum_{I,J}\delta_{p_{I3J},0}\frac{\bar{\gamma}_{s_I}^{s_1,s_2}\gamma_{s_J}^{s_I,s_3}}{\bar{z}_{13}}z_{13}^{\bar{p}_{12I}-1}B(2\bar{h}_1+2\bar{h}_2-2,2\bar{h}_3)\\
&\qquad\qquad\times\left[B(2h_1+\bar{p}_{12I}-1,2h_2+\bar{p}_{12I})-B(2h_1+\bar{p}_{12I}-1,2h_2)\right]\mathcal{O}_J(z_3,\bar{z}_3)\\
&\qquad+\sum_{I}\frac{\gamma_{s_I}^{s_1,s_2}}{z_{13}}\bar{z}_{13}^{p_{12I}}B(2\bar{h}_1+p_{12I},2\bar{h}_2+p_{12I}):\mathcal{O}_I\mathcal{O}_3:(z_3,\bar{z}_3)\\
&\qquad+\sum_{I}\frac{\bar{\gamma}_{s_I}^{s_1,s_2}}{\bar{z}_{13}}z_{13}^{\bar{p}_{12I}}B(2h_1+\bar{p}_{12I},2h_2+\bar{p}_{12I}):\mathcal{O}_I\mathcal{O}_3:(z_3,\bar{z}_3)\\
&\qquad+\sum_{I}\frac{\gamma_{s_I}^{s_1,s_3}}{z_{13}}\bar{z}_{13}^{p_{13I}}B(2\bar{h}_1+p_{13I},2\bar{h}_3+p_{13I}):\mathcal{O}_2\mathcal{O}_I:(z_3,\bar{z}_3)\\
&\qquad+\sum_{I}\frac{\bar{\gamma}_{s_I}^{s_1,s_3}}{\bar{z}_{13}}z_{13}^{\bar{p}_{13I}}B(2h_1+\bar{p}_{13I},2h_3+\bar{p}_{13I}):\mathcal{O}_2\mathcal{O}_I:(z_3,\bar{z}_3)
\end{aligned}
\tag{73}
$$

plus descendants, where p... TODO.

6 Checks Against Gravitons in the Bulk

As noted above, from the bulk scattering perspect it is not at all obvious how single- and multi-particle contributions to the composite OPE should organize themselves. For example, even in the fairly recent paper [**?**] contains the ansatz for composite negative-helicity gluons

$$
\mathcal{O}_1^{a,-}(z_1,\bar{z}_1):\mathcal{O}_2^{b,-}\mathcal{O}_3^{c,-}:(z_3,\bar{z}_3)\quad\sim\frac{\mathcal{C}^{a^-b^-c^-}{}_{d^-}}{\bar{z}_{13}^2}\mathcal{O}_p^{d,-}(z_3,\bar{z}_3)+\frac{\mathcal{D}^{a^-b^-c^-}{}_{d^-e^-}}{\bar{z}_{13}}:\mathcal{O}_p^{d,-}\mathcal{O}_q^{e,-}:(z_3,\bar{z}_3)
\tag{74}
$$

with $\mathcal{C},\mathcal{F}$ treated as independent coefficients to be determined. For tree-level Einstein gravity expanded about flat space, we are able to compute both of these contributions using bulk methods, and find that they agree exactly with the result of our complex-analytic boundary method.

6.1 Leading Single-Particle Contributions

As noted above, we already know the pairwise OPE between celestial gravitons in (matter-free tree-level perturbative about flat-space) Einstein. The authors of [**?**,**?**] use purely bulk methods to infer the leading multicollinear singularity for bulk graviton scattering, then take a Mellin transform to deduce the leading single-particle contributions to the composite OPE. Concretely, they move into spinor-helicity notation

$$
|p\rangle=\epsilon\sqrt{2\omega}\begin{bmatrix}1\\z\end{bmatrix},\quad[p]=\sqrt{2\omega}\begin{bmatrix}1\\\bar{z}\end{bmatrix}
\tag{75}
$$

and parametrize a holomorpic collinear limit as $|p_i\rangle\approx|p_j\rangle$ with $[p_i],[p_j]$ fixed. They parametrize the 3-collinear limit as

$$
z_1=z_3+\varepsilon,\quad z_2=z_2+\eta\varepsilon.
\tag{76}
$$

Then in the regime of asymptotically small ε, we can take $\eta \to \varepsilon, 1, \infty$ to determine the order in which the three limits are taken. At general η but at asymptotically small ε, an n-point graviton amplitude has leading divergence characterized by a splitting function

$$\mathcal{A}_n\left(1^{s_1}2^{s_2}3^{s_3}4^{s_4}\cdots n^{s_n}\right) = \sum_J \underbrace{\text{Split}\left(1^{s_1}2^{s_2}3^{s_3} \to J^{s_J}\right)\mathcal{A}_{n-2}\left(J^{s_J}\cdots n^{s_n}\right)}_{\sim\varepsilon^{-2}} + \mathcal{O}\left(\varepsilon^{-1}\right) \tag{77}$$

where J^{s_J} is the internal graviton that goes on-shell, and where we must sum over the two possible helicities J. The authors of [**?**] compute this splitting function by diagrammatic arguments, then Mellin-transform to find

$$\begin{aligned}
&\mathcal{O}^{++}_{\Delta_1}(z_1,\bar{z}_1)\mathcal{O}^{++}_{\Delta_2}(z_2,\bar{z}_2)\mathcal{O}^{++}_{\Delta_3}(z_3,\bar{z}_3)\\
&= \left(\frac{\kappa}{2}\right)^2 \sum_{n=0}^{\infty}\sum_{m=0}^{n} \frac{\bar{z}_{13}^m \bar{z}_{23}^{n-m}}{m!(n-m)!}\bar{\partial}^n \mathcal{O}^{++}_{\Delta_1+\Delta_2+\Delta_3}(z_3,\bar{z}_3)\Bigg[B(\Delta_1+m,\Delta_2-1+n-m,\Delta_3-1)\frac{\bar{z}_{12}\bar{z}_{13}}{z_{12}z_{13}}\\
&\quad + B(\Delta_1-1+m,\Delta_2+n-m,\Delta_3-1)\frac{\bar{z}_{12}\bar{z}_{23}}{z_{12}z_{23}} + B(\Delta_1-1+m,\Delta_2-1+n-m,\Delta_3)\frac{\bar{z}_{13}\bar{z}_{23}}{z_{13}z_{23}}\Bigg]
\end{aligned} \tag{78}$$

plus terms that contribute only at subleading order in the 3-collinear divergence. We repeat their calculations to include arbitrary combinations of graviton helicities, manipulate the relevant Laurent series to extract $\mathcal{O}_1 : \mathcal{O}_2\mathcal{O}_3 :$, and from this read off the single-particle contributions to the OPE. We find that the results agree exactly with our boundary calculation by generalized Wick starting from the pairwise OPEs.

6.2 Multiparticle Contributions and MHV Gravitons

To compute the multiparticle contributions to the composite graviton OPE, we restrict our attention to MHV amplitudes and in exchange are granted an exact recursion relation between tree-level graviton amplitudes [**?**, **?**]. In spinor-helicity notation, we have

$$\begin{aligned}
\mathcal{A}_n^{\text{MHV}} &= \mathcal{A}_n^{\text{MHV}}(\{\lambda_1,\tilde{\lambda}_1\},\ldots,\{\lambda_n,\tilde{\lambda}_n\})\\
&= \frac{\kappa}{2}\sum_{i=2}^{n-2}\frac{[1i]\langle ni\rangle\langle(n-1)i\rangle}{\langle 1i\rangle\langle n1\rangle\langle(n-1)1\rangle}\mathcal{A}_{n-1}^{\text{MHV}}\left(\{\lambda_2,\tilde{\lambda}_2\},\cdots,\left\{\lambda_i,\tilde{\lambda}_i+\frac{\langle n1\rangle}{\langle ni\rangle}\tilde{\lambda}_1\right\},\cdots,\left\{\lambda_n,\tilde{\lambda}_n+\frac{\langle i1\rangle}{\langle in\rangle}\tilde{\lambda}_1\right\}\right).
\end{aligned} \tag{79}$$

The exact calculation is involved and subtle, but we can say a few words: this relation expression an n-point amplitude in terms of an $(n-1)$pt amplitude. Applying it twice gives us an $(n-2)$pt amplitude. This is exactly the kind of relation we ought to hope to find in the bulk, as this kind of recursion in the degree of an amplitude is exactly the bulk dual of computing boundary correlators recursively in the number of correlators. The latter is in turn exactly what we do when we compute an OPE. So it is perhaps not surprising that this bulk recursion relation lets us compute not just the multiparticle contributions to the composite OPE of boundary gravitons, but the single-particle contributions as well. We find

$$\begin{aligned}
&\langle G^+_{\Delta_1}(z_1,\bar{z}_1) : G^+_{\Delta_2}G^{\pm}_{\Delta_3} : (z_3,\bar{z}_3)\cdots\rangle\\
&\quad\sim \frac{\kappa^2}{4}\frac{\bar{z}_{13}^2}{z_{13}^2}\sum_{m=0}^{\infty}\frac{\bar{z}_{13}^m}{m!}B(2\bar{h}_1+2+m,2\bar{h}_2+1,2\bar{h}_3+1)\langle\bar{\partial}^m G^{\pm}_{\Delta_1+\Delta_2+\Delta_3}(z_3,\bar{z}_3)\cdots\rangle\\
&\qquad - \frac{\kappa}{2}\frac{\bar{z}_{13}}{z_{13}}\sum_{m=0}^{\infty}\frac{\bar{z}_{13}^m}{m!}B(2\bar{h}_1+1+m,2\bar{h}_2+1)\langle:\left(\bar{\partial}^m G^+_{\Delta_1+\Delta_2}\right)G^{\pm}_{\Delta_3} : (z_3,\bar{z}_3)\cdots\rangle\\
&\qquad - \frac{\kappa}{2}\frac{\bar{z}_{13}}{z_{13}}\sum_{m=0}^{\infty}\frac{\bar{z}_{13}^m}{m!}B(2\bar{h}_1+1+m,2\bar{h}_3+1)\langle: G^+_{\Delta_2}\bar{\partial}^m G^{\pm}_{\Delta_1+\Delta_3} : (z_3,\bar{z}_3)\cdots\rangle,
\end{aligned} \tag{80}$$

in exact agreement with our boundary result.

7 OPEs from Bulk Poincaré

Now we return to a story we alluded to earlier, wherein the structure of an OPE can be largely fixed from symmetry considerations. Consider a general celestial OPE between global conformal primaries that is consistent with boundary scaling, rotations and translations

$$\mathcal{O}_{h_1,\bar{h}_1}(z,\bar{z})\mathcal{O}_{h_2,\bar{h}_2}(0,0) = \frac{1}{z}\sum_{m=0}^{\infty} C_p^{(m)}(\bar{h}_1,\bar{h}_2)\bar{z}^{p+m}\bar{\partial}^m \mathcal{O}_{h_1+h_2-1,\bar{h}_1+\bar{h}_2+p}(0,0) + \mathcal{O}\left(z^0\right). \tag{81}$$

Here we write $\mathcal{O}_{h_i,\bar{h}_i}$ rather than $\mathcal{O}_i$ to be slightly more explicit. The operator $\mathcal{O}_{h_1+h_2-1,\bar{h}_1+\bar{h}_2+p}(0,0)$ is a global conformal primary, and a sum over conformal families is implied (by linearity, for our argument it will suffice to consider just a single primary). p is just a useful notation for the shift in weights, and we can write variously as

$$p = \bar{h}_P - \bar{h}_1 - \bar{h}_2 = s_1 + s_2 - s_P - 1. \tag{82}$$

The coefficients depend in general on $(h_1, h_2, \bar{h}_1, \bar{h}_2)$, but we think of them equivalently as functions of $(s_1, s_2, \bar{h}_1, \bar{h}_2)$ and suppress the spin-dependence in our notation.

Now this ansatz for our OPE already enforces covariance under most generators of the boundary conformal group, but does not yet enforce covariance under special conformal transformations. To get this, we recall (or deduce from the OPE against the stress tensor) that the antiholomorphic generator of special conformal transformations acts on conformal primaries as

$$\begin{aligned} \left[\bar{L}_n, \mathcal{O}_{h,\bar{h}}(z,\bar{z})\right] &= \bar{z}^n\left((n+1)\bar{h} + \bar{z}\bar{\partial}\right)\mathcal{O}_{h,\bar{h}}(z,\bar{z}) \\ \implies \left[\bar{L}_1, \mathcal{O}_{h,\bar{h}}(z,\bar{z})\right] &= \bar{z}\left(2\bar{h} + \bar{z}\bar{\partial}\right)\mathcal{O}_{h,\bar{h}}(z,\bar{z}). \end{aligned} \tag{83}$$

As was shown in [**?**], applying $\left[\bar{L}_1, \cdot\right]$ this to both sides of our ansatz (81) and equating the results term by term gives the recursion relation for the OPE coefficients

$$\left(2\bar{h}_1 + p + m\right) C_p^{(m)}(\bar{h}_1,\bar{h}_2) = (m+1)(2\bar{h}_1 + 2\bar{h}_2 + 2p + m)C_p^{(m+1)}(\bar{h}_1,\bar{h}_2). \tag{84}$$

Observe that the spins play no role here. This lets us recursively solve for $C^{(m+1)}$ in terms of $C^{(m)}$, but we can do better. Bulk Lorentz symmetry is embedded inside of bulk Poincaré, and covariance under bulk translations has a celestial image acting as

$$\left[P_{m,n}, \mathcal{O}_{h,\bar{h}}(z,\bar{z})\right] = \frac{1}{2}z^{m+\frac{1}{2}}\bar{z}^{n+\frac{1}{2}}\mathcal{O}_{h+\frac{1}{2},\bar{h}+\frac{1}{2}}(z,\bar{z}), \quad m,n = \pm\frac{1}{2}. \tag{85}$$

Together with the global conformal generators these form a closed algebra

$$\begin{aligned} [L_k, L_\ell] &= (k-\ell)L_{k+\ell}, \\ [L_k, P_{m,n}] &= \left(\frac{k}{2} - m\right)P_{m+k,n}. \end{aligned} \tag{86}$$

If we play the same game with $\left[P_{-\frac{1}{2},-\frac{1}{2}}, \cdot\right]$, applying it to both sides of (81) and equating the results term-by-term, we can the fixed-m constraint [**?**]

$$C_p^{(m)}\left(\bar{h}_1 + \frac{1}{2}, \bar{h}_2\right) + C_p^{(m)}\left(\bar{h}_1, \bar{h}_2 + \frac{1}{2}\right) = C_p^{(m)}\left(\bar{h}_1, \bar{h}_2\right). \tag{87}$$

Repeating this logic with $P_{-\frac{1}{2},\frac{1}{2}}$ gives

$$C_p^{(m)}\left(\bar{h}_1 + \frac{1}{2}, \bar{h}_2\right) = (m+1)C_p^{(m+1)}\left(\bar{h}_1, \bar{h}_2\right). \tag{88}$$

Taking appropriate linear combinations of these constraints at $m = 0$ fixes $C_p^{(0)}(\bar{h}_1,\bar{h}_2)$, and subsequently applying the recursive constraint in m (84) fixes the general-m coefficient to take the form

$$C_p^{(m)}(\bar{h}_1,\bar{h}_2) = \frac{\gamma_{s_I}^{s_1,s_2}}{m!}B(2\bar{h}_1 + p + m, 2\bar{h}_2 + p), \tag{89}$$

where $s_I = s_1 + s_2 - p - 1$. So by purely representation-theoretic means we can fix the OPE structure up to a single leading constant depending on the involves spins, which can be thought of as essentially the coupling constant for the corresponding bulk 3pt vertex.

In the work [**?**], we generalize this game. We work out the action of bulk Poincaré on composite operators with derivative insertions, take a more general ansatz

$$\begin{aligned}
&\mathcal{O}_{h_1,\bar{h}_1}(z,\bar{z}) : \mathcal{O}_{h_2,\bar{h}_2}\mathcal{O}_{h_3,\bar{h}_3} : (0,0) \\
&= \sum_{p\geq 0}\sum_{m=0}^{\infty} C_p^{(m)}(\bar{h}_1,\bar{h}_2,\bar{h}_3)\frac{\bar{z}^{p+m}}{z^2}\bar{\partial}^m \mathcal{O}_{h_1+h_2+h_3-2,\bar{h}_1+\bar{h}_2+\bar{h}_3+p}(0,0) \\
&\quad + \sum_{p\geq 0}\sum_{m=0}^{\infty} D_p^{(m)}(\bar{h}_1,\bar{h}_2,\bar{h}_3)\frac{\bar{z}^{p+m}}{z} : \left(\bar{\partial}^m \mathcal{O}_{h_1+h_2-1,\bar{h}_1+\bar{h}_2+p}\right)\mathcal{O}_{h_3,\bar{h}_3} : (0,0) \\
&\quad + \sum_{p\geq 0}\sum_{m=0}^{\infty} E_p^{(m)}(\bar{h}_1,\bar{h}_2,\bar{h}_3)\frac{\bar{z}^{p+m}}{z} : \mathcal{O}_{h_2,\bar{h}_2}\bar{\partial}^m \mathcal{O}_{h_1+h_3-1,\bar{h}_1+\bar{h}_3+p} : (0,0),
\end{aligned} \tag{90}$$

and similarly demand covariance term-by-term. We find that the result gives exactly the same result as our integral prescription (69).

8 It's just Complex Analysis!

At the beginning of our work on [**?**], we thought of the checks above (compute multicollinear limits in the bulk, then on the boundary, and compare) as non-trivial checks of the celestial holography. After all, as we emphasized above the holographic dictionary is in some sense purely representation-theoretic, and it doesn't manifestly lead to a local boundary theory. However, in the course of the work we realized we were thinking about things incorrectly: the generalized Wick theorem (69) is not a non-trivial fact about locality properties of 2D CFTs; it is purely a fact about complex analysis in several variables. Concretely, consider the function of three complex variables

$$f(z_1,z_2,z_3) = \frac{1}{z_{12}^A z_{13}^B z_{23}^C} \text{ for } A,B,C \in \mathbb{Z}_{>0}. \tag{91}$$

. We can play the same games here as we did for pairwise OPEs. We defined the $:23:$ normal-ordered of f by the prescription

- Decompose $z_{12} = z_{13} - z_{23}$ so that f becomes a function of z_{23}, z_{13}.
- Thinking of the latter as independent variables, expand as a Laurent series in z_{23}.
- Extract the term of order exactly z_{23}^0.

For our exemplary function we find

$$f(z_1,z_2,z_3)\Big|_{\Theta(z_{23}^0)} = \binom{A+C-1}{C}\frac{1}{z_{13}^{A+B+C}}. \tag{92}$$

We define the Wick contraction $\overset{\sqcap}{12}$ of f to be the singular part in the Laurent series in z_{12} after expanding $z_{13} = z_{12} + z_{23}$, and similarly for $\overset{\sqcap}{13}$. From these we find

$$\begin{aligned}
f(z_1,z_2,z_3)\Big|_{\text{singular in } z_{12}} &= \frac{1}{z_{12}^A z_{23}^{B+C}}\sum_{k=0}^{A}(-1)^k\binom{B+k-1}{k}\left(\frac{z_{12}}{z_{23}}\right)^k, \\
f(z_1,z_2,z_3)\Big|_{\text{singular in } z_{13}} &= \frac{(-1)^A}{z_{13}^B z_{23}^{A+C}}\sum_{k=0}^{B}\binom{A+k-1}{k}\left(\frac{z_{13}}{z_{23}}\right)^k.
\end{aligned} \tag{93}$$

Then by a direct calculation, we find that the generalized Wick theorem

$$f(z_1,z_2,z_3)\Big|_{\Theta(z_{23}^0)}\Big|_{\text{singular in } z_{13}} = \oint_{z_3}\frac{dz_2}{2\pi i}\frac{1}{z_{23}}\left[f(z_1,z_2,z_3)\Big|_{\text{singular in } z_{12}} + f(z_1,z_2,z_3)\Big|_{\text{singular in } z_{13}}\right] \tag{94}$$

holds exactly. Morally, there is no essential field-theoretic content in the generalized Wick theorem (69), only algebraic facts about Laurent series in multiple variables. So our bulk:boundary checks could not possibly have failed.

9 TODO

Explain why bulk physics constraints Δ_3. Mention in particular that this is why the weights conspire to enforce integer powers in the OPE.